

Selection

Evolution & Ecology

Jacopo Grilli, Ciudad de Guatemala, 27/11/2025

Plan

~~L1 - Exponential growth~~

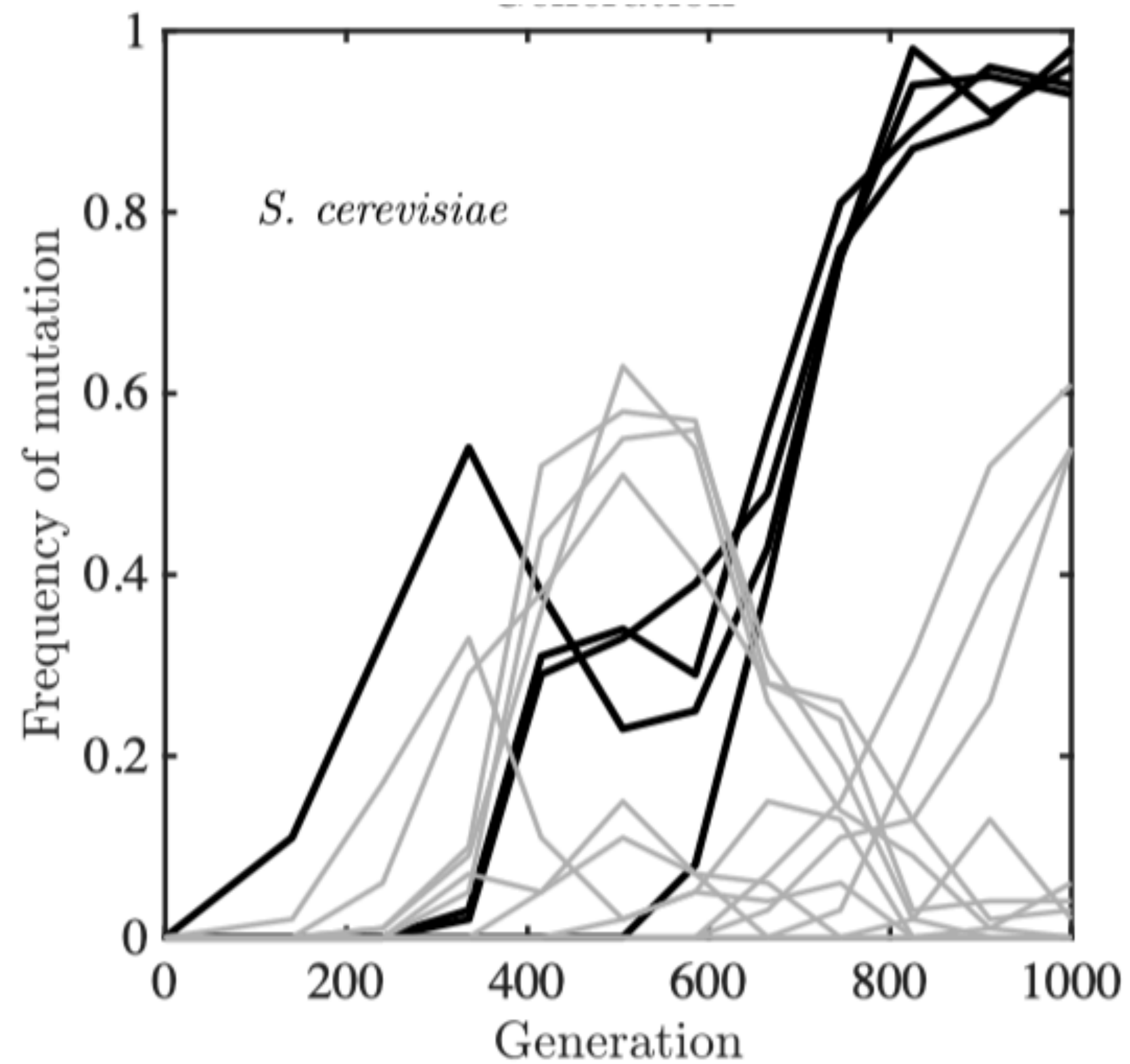
~~L2 - Mutations~~

L3 - Selection

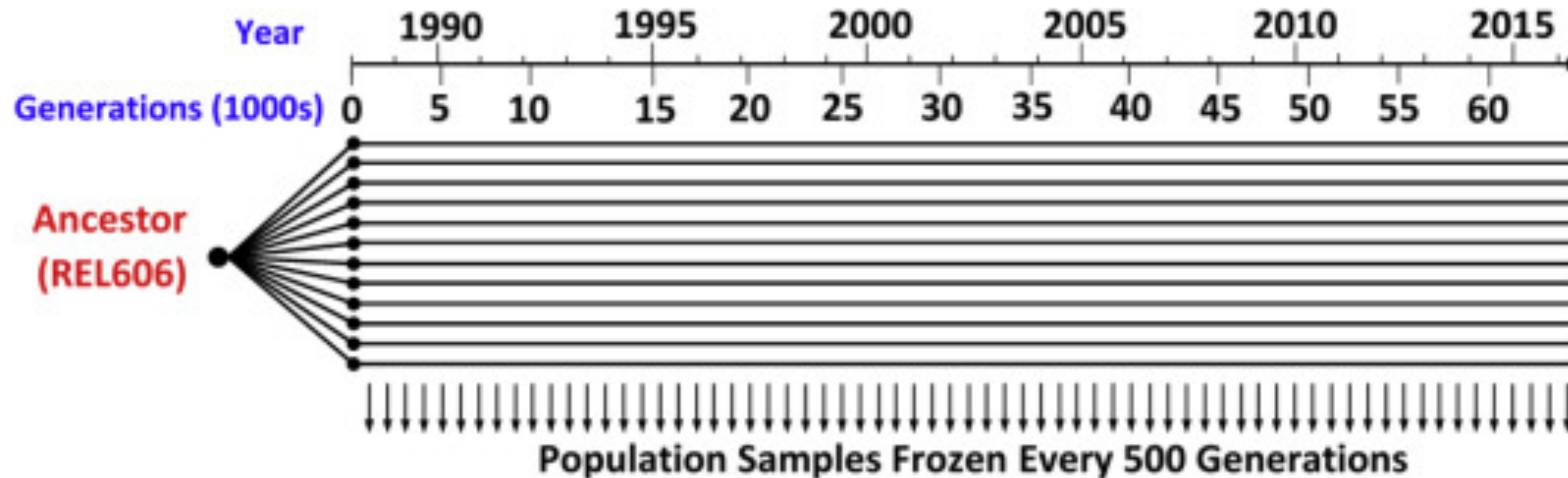
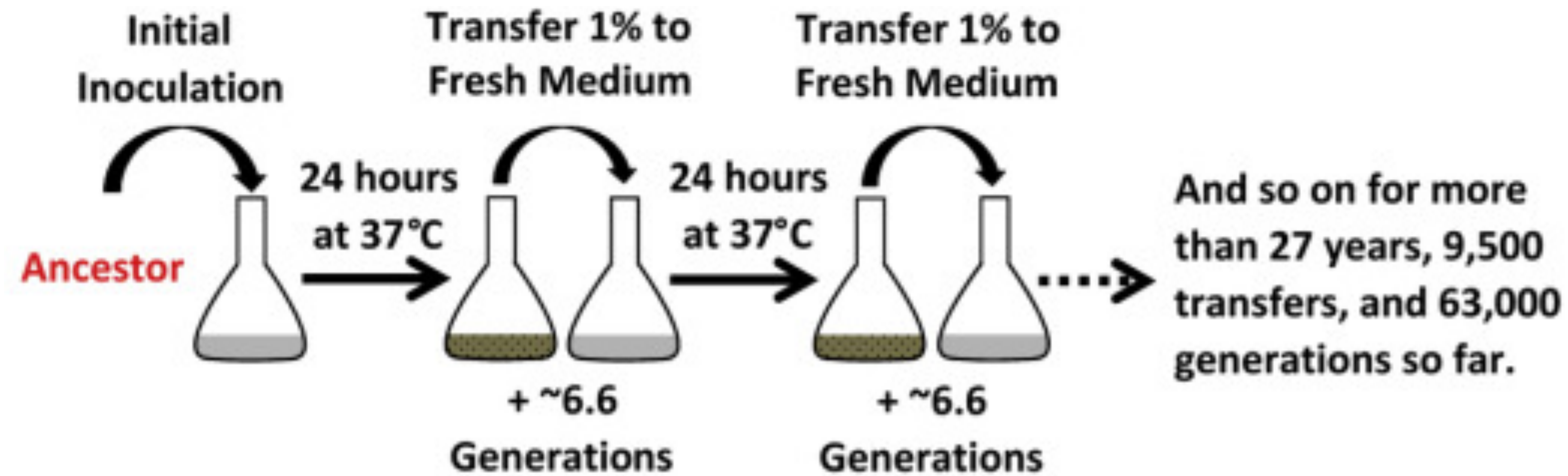
L4 - Ecology

L5 - Eco-Evo

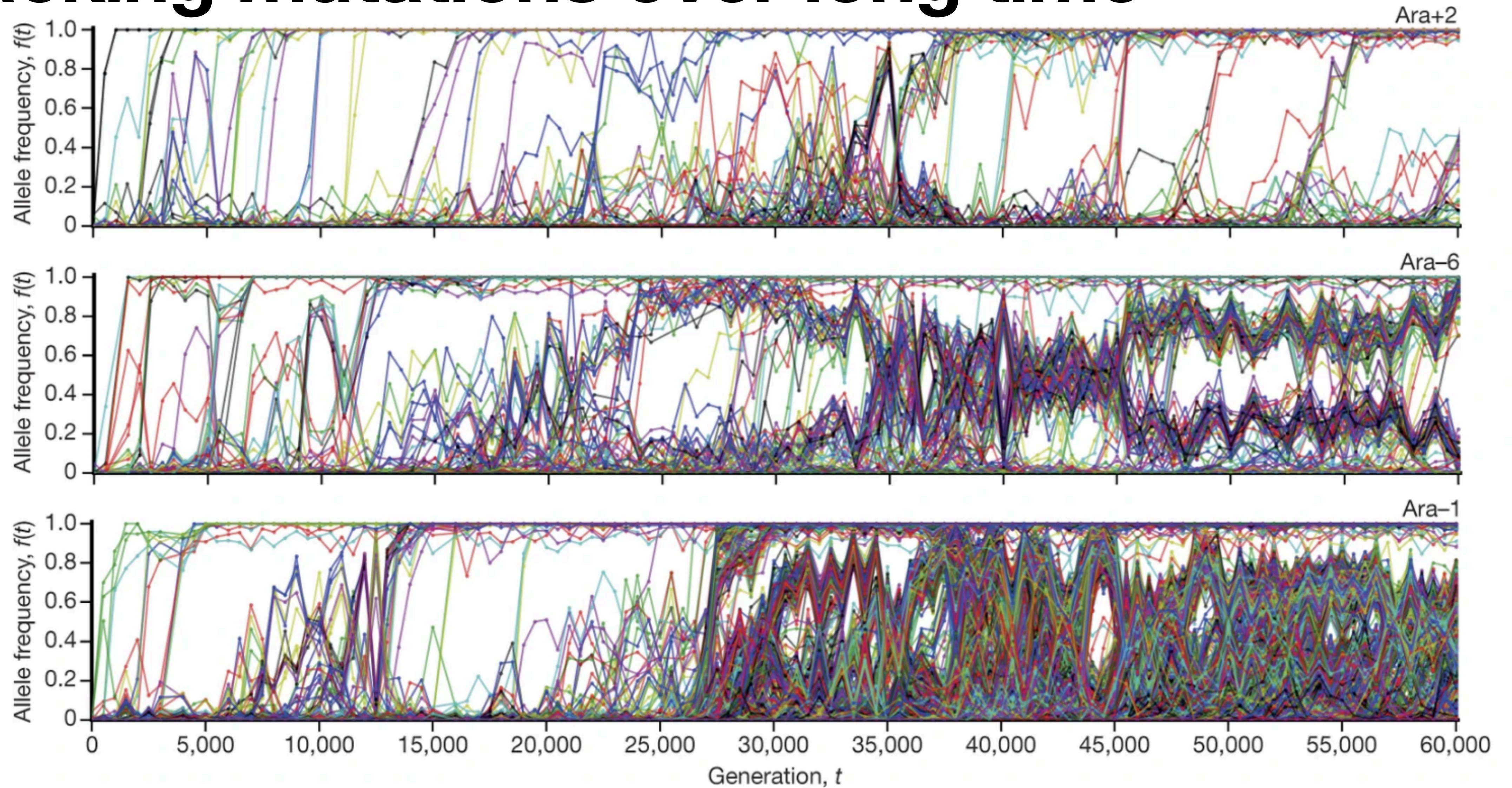
Evolution in the lab, tracking mutations



Long-term Evolution Experiment



Tracking mutations over long time



Good et. al, Nature 2017

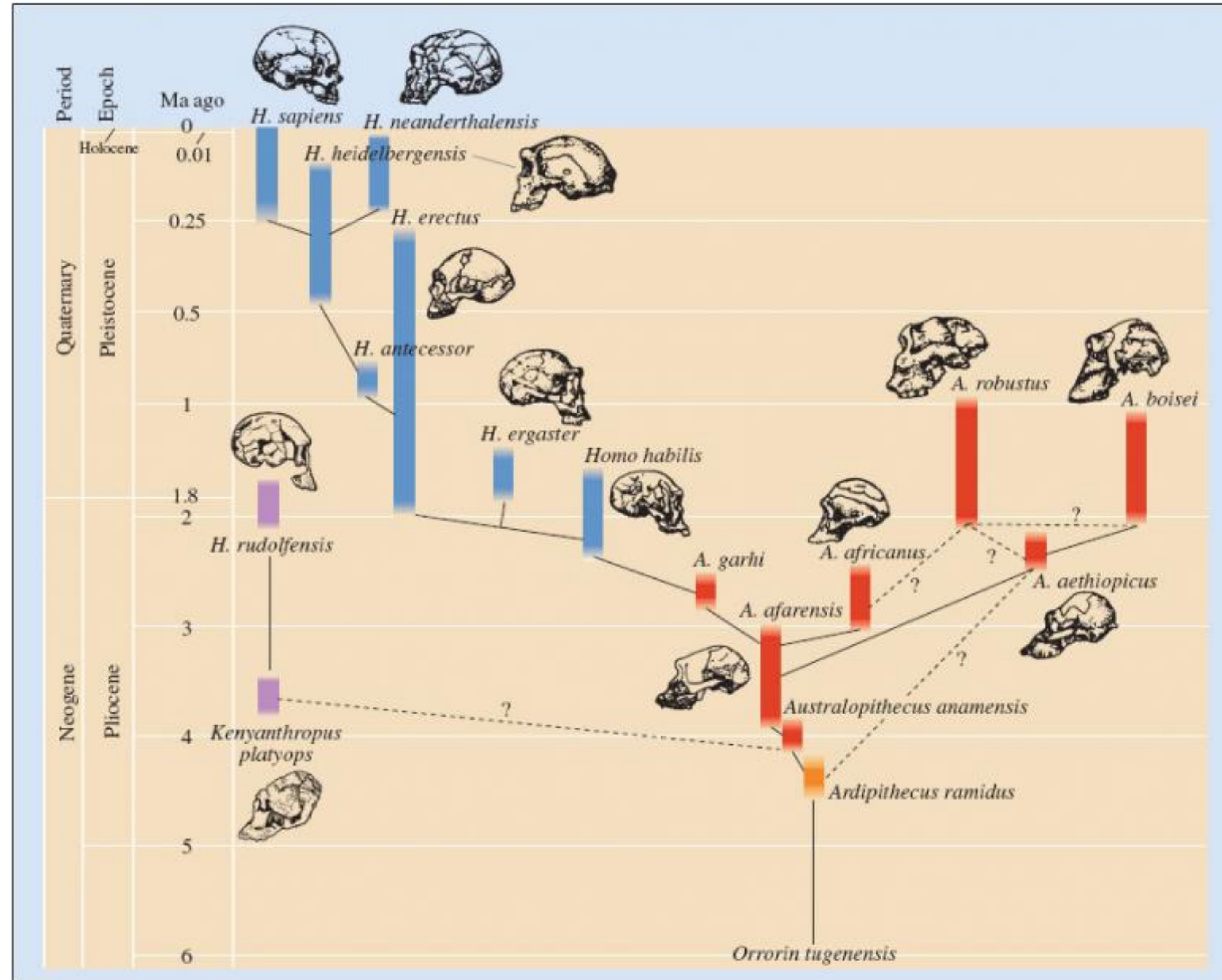
Long-term is really long-term

75000 generations

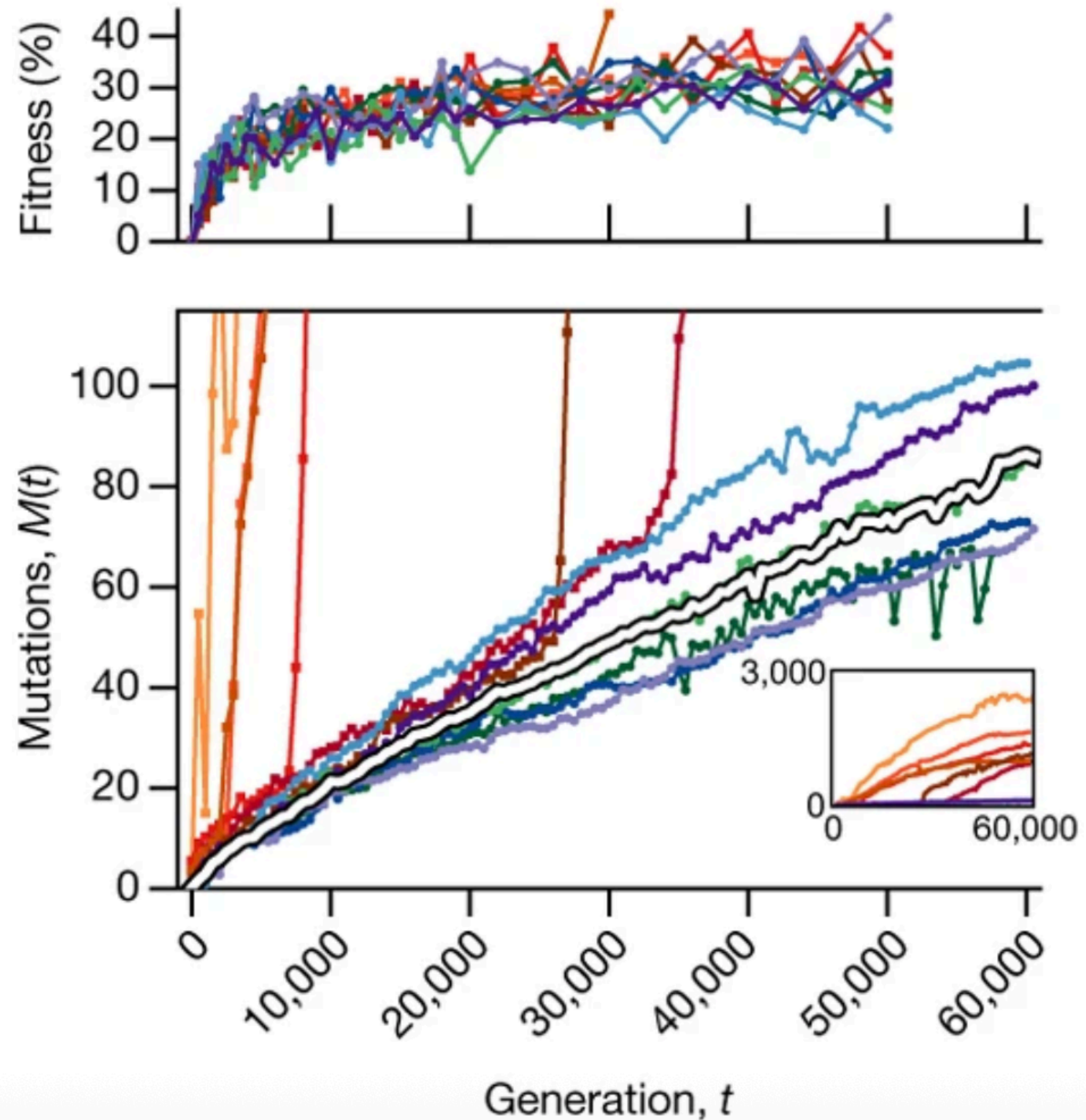
human generation ~ 20 years

~1.5M years ago

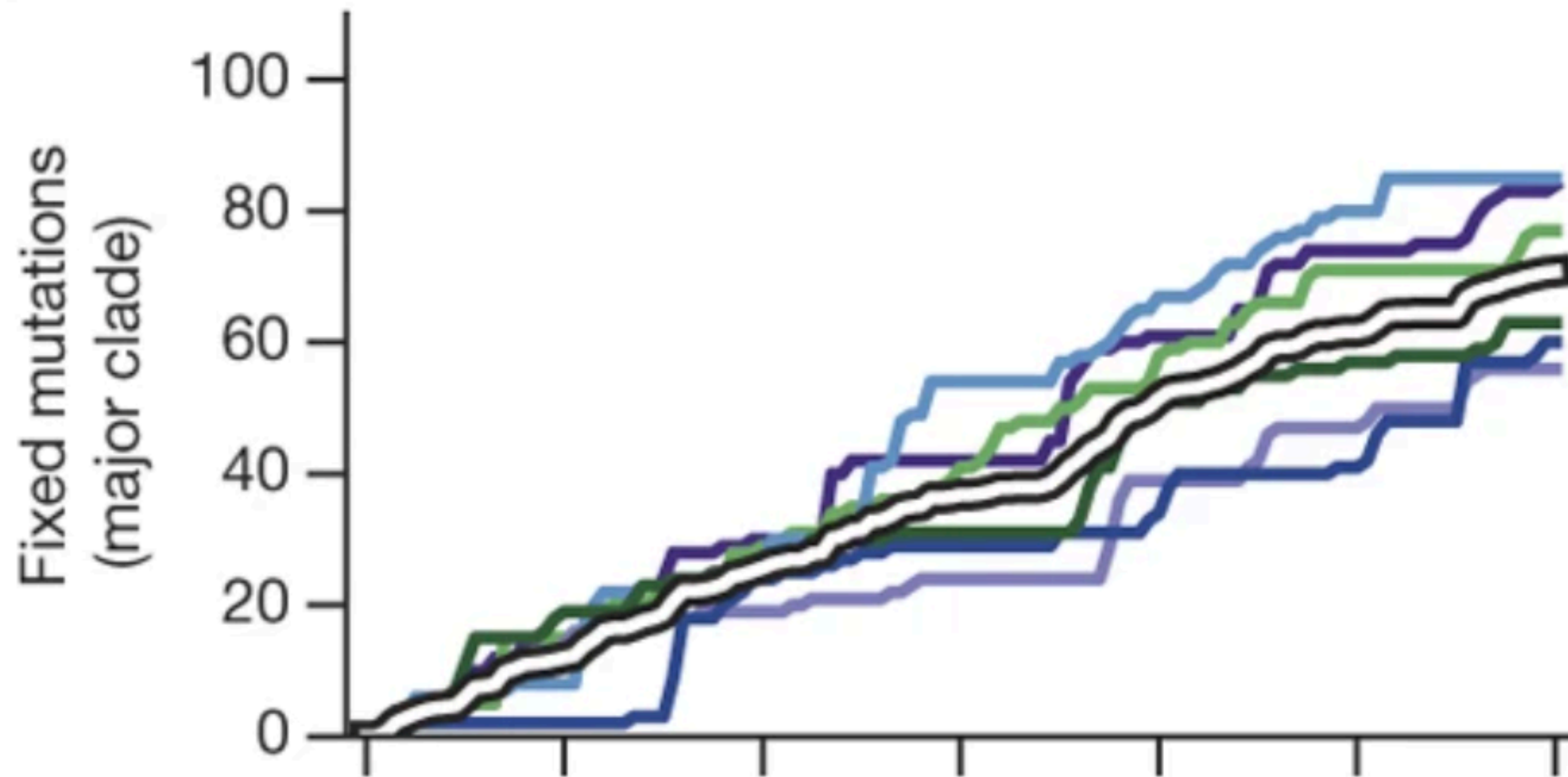
appearance of the genus Homo



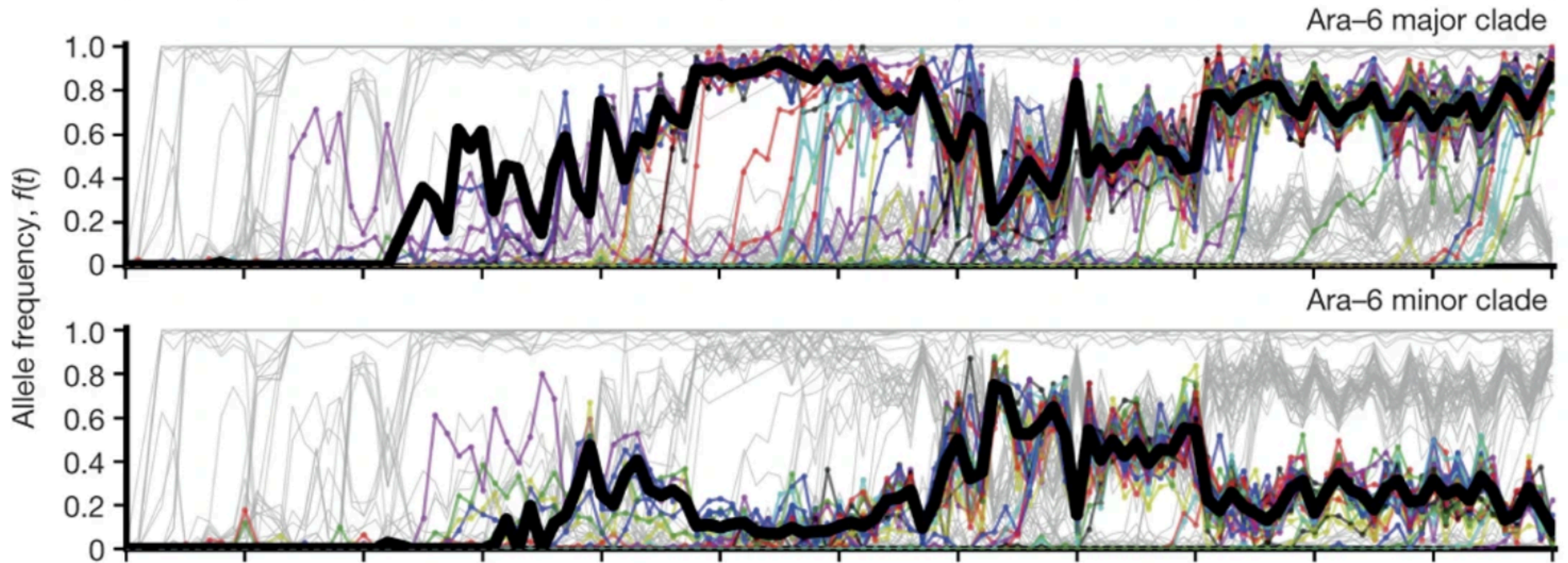
Mutations have an effect



Evolution is not continuous

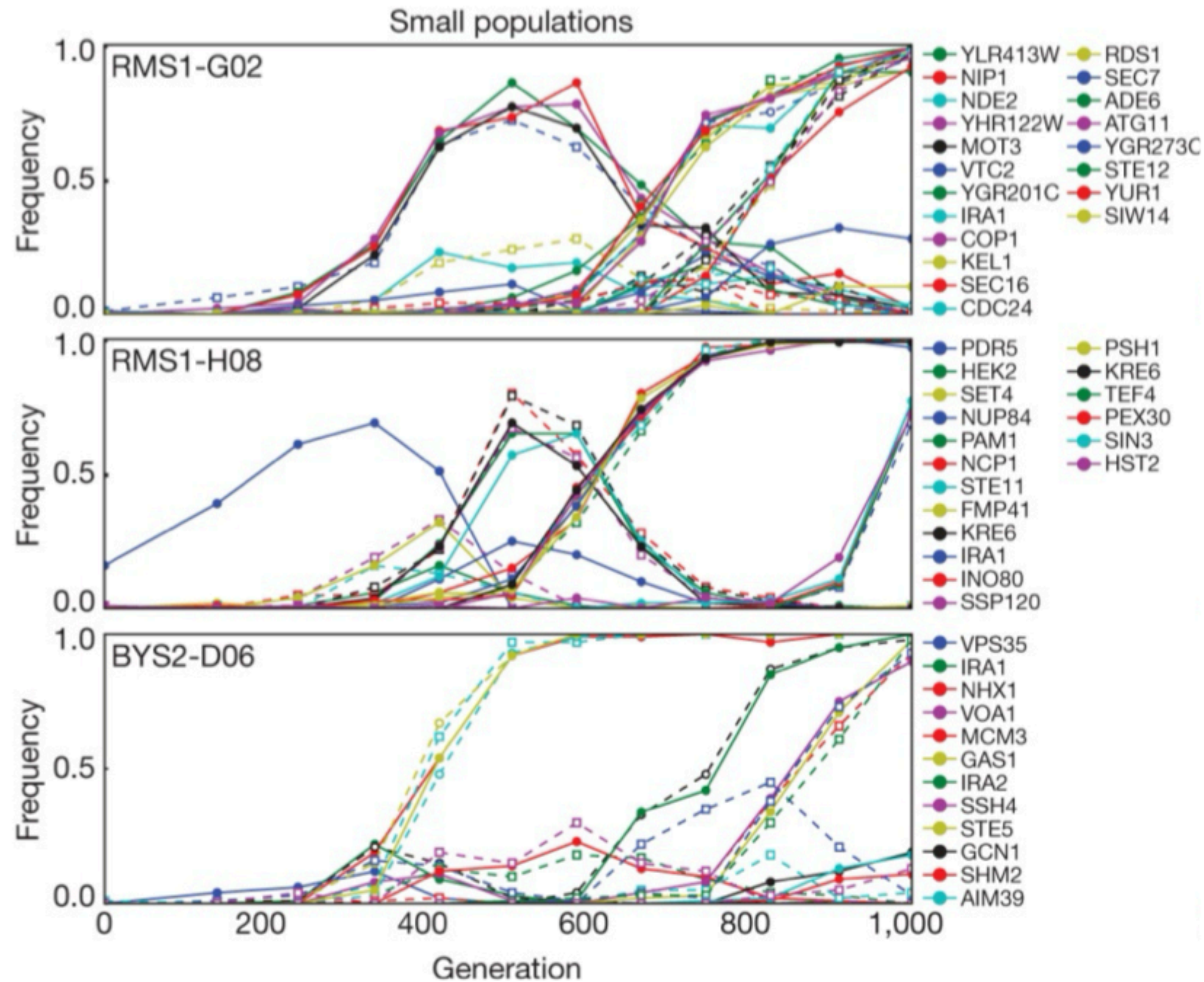


Evolution creates Ecology



A simple deterministic view

Mutations hitchhike and interfere



The role of genetic drift

Fixation probability

